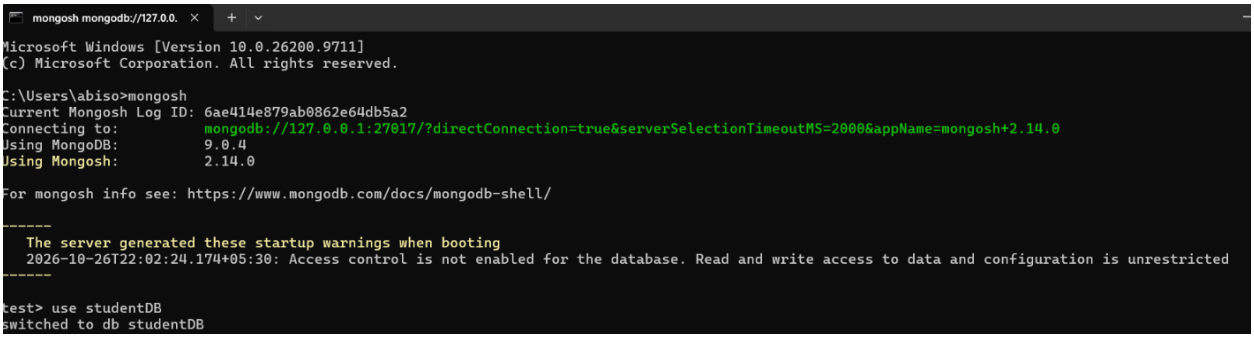
MongoDB

Database Task & Command Documentation

1. Database & Collection Setup



2. Insert Operations

```
studentDB> db.createCollection("students")
{ ok: 1 }
studentDB> db.students.insertMany([
|   {name: "Abi", age: 21, course: "MERN Stack", status: "ongoing"},
|   {name: "Michael", age: 22, course: "Python", status: "ongoing"},
|   {name: "Catherine", age: 19, course: "MERN Stack", status: "completed"},
|   {name: "Priscilla", age: 24, course: "Java", status: "ongoing"}
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ae416d479ab0862e64db5a3'),
    '1': ObjectId('6ae416d479ab0862e64db5a4'),
    '2': ObjectId('6ae416d479ab0862e64db5a5'),
    '3': ObjectId('6ae416d479ab0862e64db5a6')
  }
}
```

3. Read Operations — Fetch All Documents

```
studentDB> db.students.find()
[
  {
    _id: ObjectId('6ae185418cffd925d62cb675'),
    name: 'abi',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb676'),
    name: 'michael',
    age: 22,
    course: 'MERN Stack',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb677'),
    name: 'priscilla',
    age: 19,
    course: 'Python',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb679'),
    name: 'soniya',
    age: 21,
    course: 'MERN Stack',
    status: 'in-progress',
    scholarship: true
  },
```

4. Read Operations — Filter Documents

```
studentDB> db.students.find({ age:{$in:[20,22,25]} })
[
  {
    _id: ObjectId('6ae185418cffd925d62cb675'),
    name: 'abi',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb676'),
    name: 'michael',
    age: 22,
    course: 'MERN Stack',
    status: 'in-progress'
  }
]
studentDB>
```

5. Update Operations

[Insert Screenshot Her

```
studentDB> db.students.updateOne({name:"Abi"},{$set:{age:25}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
studentDB>
```

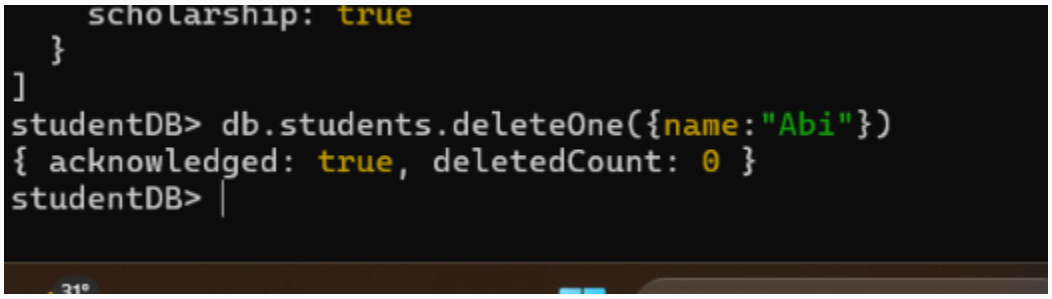
e]

6. Query Operator — $gt

```
studentDB> db.students.find({age:{$gt:20}})
[
  {
    _id: ObjectId('6ae185418cffd925d62cb676'),
    name: 'michael',
    age: 22,
    course: 'MERN Stack',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb679'),
    name: 'soniya',
    age: 21,
    course: 'MERN Stack',
    status: 'in-progress',
    scholarship: true
  }
]
```

7. Delete Operations

[Insert Screenshot



Here]

8. create a collection

[Insert Screenshot



Here]

9. insert one document into students collection



10. insert multiple collection

```
studentDB> db.students.insertMany([
|       {
|           name: "Michael",
|           age: 23,
|           course: "MERN Stack",
|           status: "Active"
|       },
|       {
|           name: "Catherine",
|           age: 22,
|           course: "Python",
|           status: "Active"
|       },
|       {
|           name: "Priscilla",
|           age: 24,
|           course: "Java",
|           status: "Inactive"
|       },
|       {
|           name: "Abi",
|           age: 21,
|           course: "Full Stack Development",
|           status: "Active"
|       }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6ae42fd279ab0862e64db5ad'),
    '1': ObjectId('6ae42fd279ab0862e64db5ae'),
    '2': ObjectId('6ae42fd279ab0862e64db5af'),
    '3': ObjectId('6ae42fd279ab0862e64db5b0')
  }
}
```

11. fetch all documents from collection

```
}
studentDB> db.students.find()
[
  {
    _id: ObjectId('6ae185418cffd925d62cb675'),
    name: 'abi',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb676'),
    name: 'michael',
    age: 22,
    course: 'MERN Stack',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb677'),
    name: 'priscilla',
    age: 19,
    course: 'Python',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb679'),
    name: 'soniya',
    age: 21,
    course: 'MERN Stack',
    status: 'in-progress',
    scholarship: true
  },
  {
    _id: ObjectId('6ae42e7c79ab0862e64db5ac'),
    studentId: 101,
    name: 'Abi',
    age: 22,
    course: 'Full Stack Development',
    city: 'Chennai',
    status: 'Active'
  },
```

12. use filter to fetch specific documents

```
studentDB> db.students.find({name:"Michael"})
[
  {
    _id: ObjectId('6ae42fd279ab0862e64db5ad'),
    name: 'Michael',
    age: 23,
    course: 'MERN Stack',
    status: 'Active'
  }
]
studentDB>
```

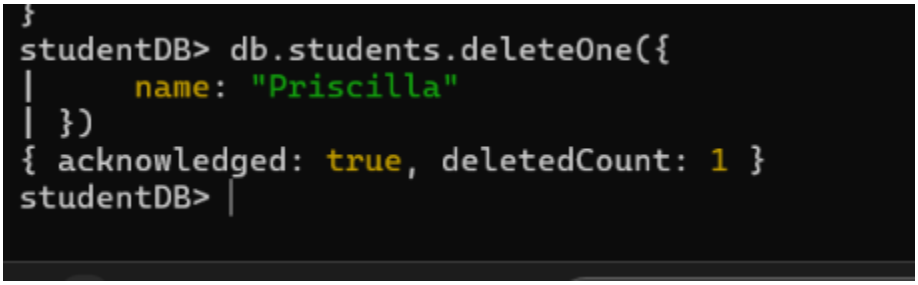
13. update a single documents

```
studentDB> db.students.updateOne(
|      { name: "Michael" },
|      {
|          $set: {
|              age: 25,
|              status: "Completed"
|          }
|      }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB>
```

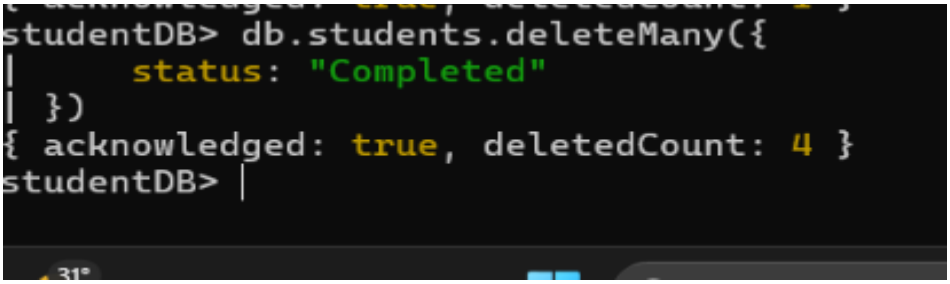
14. update a multiple documents

```
studentDB> db.students.updateMany(
|      { status: "Active" },
|      {
|          $set: {
|              status: "Completed"
|          }
|      }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 3,
  modifiedCount: 3,
  upsertedCount: 0
}
studentDB>
```

15. delete one document based on the condition



16. delete multiple document based on the condition



17. $gt (greater than 20)

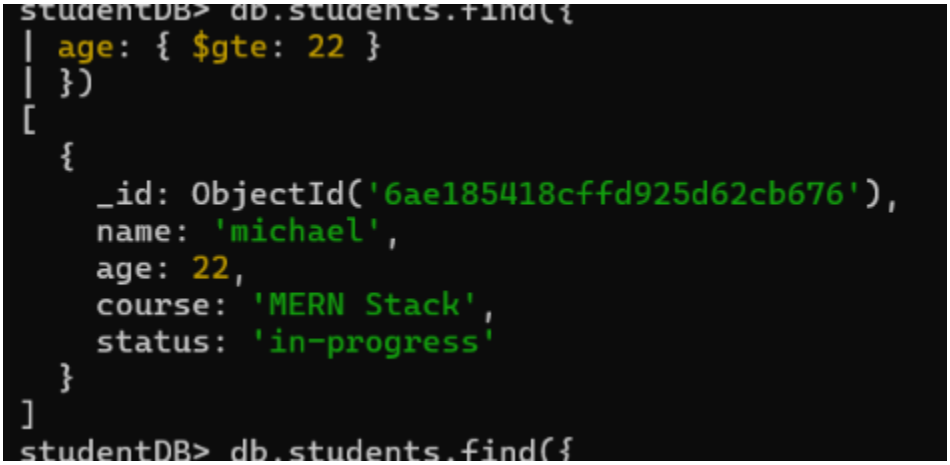
```
studentDB> db.students.find({
| age: { $gt: 22 }
| })

studentDB> db.students.find({
| age: { $lt: 22 }
| })
[
  {
    _id: ObjectId('6ae185418cffd925d62cb675'),
    name: 'abi',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb677'),
    name: 'priscilla',
    age: 19,
    course: 'Python',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb679'),
    name: 'soniya',
    age: 21,
    course: 'MERN Stack',
    status: 'in-progress',
    scholarship: true
  }
]
```

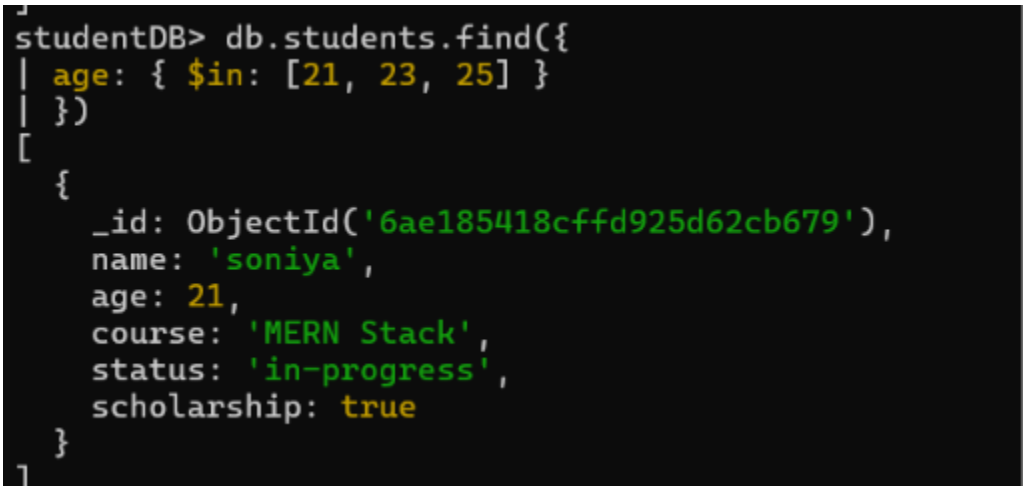
19. $lt (less than 20)

```
studentDB> db.students.find({
| age: { $lt: 22 }
| })
[
  {
    _id: ObjectId('6ae185418cffd925d62cb675'),
    name: 'abi',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb677'),
    name: 'priscilla',
    age: 19,
    course: 'Python',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb679'),
    name: 'soniya',
    age: 21,
    course: 'MERN Stack',
    status: 'in-progress',
    scholarship: true
  }
```

20. $gte (greater than or equal)



21. $in (matches any given value)



22. $nin (Excludes given values)

```
studentDB> db.students.find({
| age: { $nin: [21, 23, 25] }
| })
[
  {
    _id: ObjectId('6ae185418cffd925d62cb675'),
    name: 'abi',
    age: 20,
    course: 'MERN Stack',
    status: 'completed'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb676'),
    name: 'michael',
    age: 22,
    course: 'MERN Stack',
    status: 'in-progress'
  },
  {
    _id: ObjectId('6ae185418cffd925d62cb677'),
    name: 'priscilla',
    age: 19,
    course: 'Python',
    status: 'in-progress'
  }
```